

Conflicts of Interest in Municipal Bond Advising and Underwriting

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When can financial advisor conflicts of interest generate worse outcomes for clients? A regulation following from Dodd-Frank prohibits municipal advisors from simultaneously acting as bond underwriters. Using a difference-in-differences approach and 20,051 bond auctions, I test whether limited advisor privileges affect financial advice and borrower outcomes. Financing costs of bonds with potential dual advisor-underwriters fall by 11.4 basis points after the advisor is no longer allowed to underwrite. The decline follows from increases in standardization, third-party certification, and auction participation, all of which are consistent with limiting the adverse selection that arises from advisors withholding information from the market. (*JEL* D44, D53, G12, G14, G28, H74)

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Conflicts of interest are common in financial services, but their presence is not necessarily detrimental to customers in a market that can discipline those conflicts. A fundamental source of financial conflicts of interest is that customers purchasing financial products are less informed about the market for those products than the firms that sell them (Campbell 2016). Financial advisors are intermediaries that play a key role in providing information to customers about the market and potentially to the market about the customer. However, that role in information provision may be subject to a conflict of interest if the advisor can profit by withholding information. While recent literature has found reputational concerns and adjustments by customers can

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both help the market discipline potential conflicts,¹ there is still a widespread belief among regulators that conflicts can harm customers, including borrowers in the \$4 trillion municipal bond market in the United States (U.S. Securities and Exchange Commission 2012). Using auction data, I show how conflicts of interest among municipal financial advisors led to higher borrowing costs for states and local governments by exacerbating informational asymmetries leading to decreased market participation.

Municipal advisors are one of the most important intermediaries in the municipal bond market and provide services including structuring bonds, developing plans for how to spend and invest funds, preparing public documents, and soliciting credit rating agencies on behalf of states and local governments. At the same time, advisors have a *conflict of interest* when they are part of a firm that is also vying to provide underwriting services, which is “a situation in which a party to a transaction can potentially gain by taking actions that adversely affect its counterparty” (Mehran and Stulz 2007, p. 268). Such “dual advisors,” which made up 25% of the advising market before 2011, have a profit function that is increasing in primary market municipal borrowing costs while holding all else constant.² I show that a regulation prohibiting dual advisors from concurrently underwriting municipal debt lowers borrowing costs on average, implying that market forces alone do not fully discipline advisor conflicts of interest.

The primary market for U.S. municipal bonds provides a prime laboratory to learn about why, how, and how much conflicts of interest can impact customer outcomes for two particular reasons. First, many municipal bonds are required to be placed with an underwriter using an auction, the results of which are public. Directly observing submitted bids allows me to measure competitive forces that may discipline conflicts in a more granular way than through broad notions of market concentration or conduct. Second, the municipal bond market is made up of over 50,000 heterogeneous borrowers and segmented by state geography (Butler 2008), taxation (Babina et al. 2021), and regulation (Cestau et al. 2021). This market, which includes borrowers ranging from the state of California to school districts in rural Pennsylvania, has scope for informational asymmetries and market power that may be important for how the market disciplines potential conflicts. I find that the impact of removing a conflict of interest is largest for opaque issues where underwriting auction participation is scarce and where financial advisors are able to guide borrowers away from

¹ Mehran and Stulz (2007) provide a recent overview of the empirical literature on conflicts of interest in financial markets, focusing on sell-side analysts. They argue that the real impacts of conflicts of interest are usually benign in the empirical literature. Conflicts matter, however, in some contexts, for example, independence of mutual fund boards.

² Underwriter profits in this context are an equilibrium outcome of two markets: primary market auctions and sales to secondary market investors. All else equal, underwriter profit is increasing in the yield from the auction because it pays a lower price to the municipal issuer and is able to receive the same price from secondary market investors. If the conflicted advisor can take actions that raise the yield in both markets, it can increase underwriting profits as long as the yield increases more in the primary market than in the secondary market.

getting third-party certifications like credit ratings. Meanwhile, large portions of the market show limited impacts of the conflict or its removal.

The regulation limiting advisors' ability to underwrite was enacted by the Municipal Securities Rulemaking Board (MSRB) updating Rule G-23 as a consequence of the Dodd-Frank Wall Street Reform and Consumer Protection Act of 2010.³ [Schapiro \(2010\)](#) and [SEC \(2011\)](#) explain the reasoning behind prohibiting dual advisors from actively performing both roles: since underwriters can profit from higher interest costs in the primary market, the advice provided by an advisor seeking to act as underwriter may be biased toward higher interest cost debt structures. Critics of the regulation worried that interest costs would increase as municipal advisors are prohibited from underwriting because the underwriter market suffers from a fundamental lack of participation such that the marginal participant can be very important ([Bond Dealers of America 2019](#)). How a conflict of interest for financial advisors manifests, and whether it has any deleterious effects for clients, is ultimately an empirical question.

A financial transaction with advice can be thought of as taking part in two stages. In the first stage a financial advisor and a municipality work together to design a financial product that suits the municipality's needs taking into account a second stage in which the municipality finds a financial service firm from whom to buy the product. The extent to which a conflict of interest will affect the first stage outcome is unclear. A dual advisor can extract transfers from their municipal clients by taking any action that increases their own profits in the second stage. However, the advice may not necessarily be biased by the conflict of interest because financial advisors face potential reputational and other concerns that discipline their behavior ([Mehran and Stulz 2007](#)).

The second stage of issuing a municipal bond is often required to take place in a first-price, sealed-bid auction. Underwriters submit yields that they are willing to receive to underwrite the bond package that an issuer already structured with the help of an advisor and the low bid wins. The general theory of auctions developed by [Milgrom and Weber \(1982\)](#) provides some structure for what could happen when the advisor is removed from the second stage. In the [Milgrom and Weber \(1982\)](#) model with affiliated values, where the value of winning the auction to each participant may depend on the values of other market participants, two potential forces are at work when one competitor is added to the market. The first is the *competitive effect*: one more competitor means you need to bid a little bit more aggressively in order to win the auction and so you bid a lower yield. The second is the *adverse selection effect*. The

³ Subchapter H of Title IX of Dodd-Frank, formalized by an update to MSRB Rule G-23 on November 27, 2011, disallows advisors from vying for underwriting business for any bond issue for which they offered advice. For an overview of all of the regulatory changes affecting municipal advice since Dodd-Frank, see [Bergstresser and Luby \(2018\)](#). I abstract away from the registration and fiduciary rules in 2014 and 2015, respectively, as well as the enhanced certification requirements associated with the Series 50 exam in 2017 since these regulations affect all advisors and thus will not interfere with the difference-in-differences research design.

value of winning has a common value component, so each additional bid you beat raises the likelihood that your bid was an overestimate of this common value and you bid less aggressively (i.e., bid a higher yield).⁴ When a dual advisor is barred from participating in the second stage, the remaining potential underwriters may bid less aggressively because they have a higher chance of winning conditional on the bid, but they also could bid more aggressively because the chance of winning when their estimate of the common value is overly positive has decreased. Which effect dominates is an empirical question and can be influenced by the advisor's actions in the first stage.

Using a difference-in-differences research design to identify the net effect, I compare the borrowing cost for bonds sold at auction that employ an advisor who also offers underwriting services, a potential “dual advisor,” to bonds that are issued with advice from an independent advisor—an advisor not associated with an investment bank—before and after Rule G-23 was updated to ban concurrent advising and underwriting on November 27, 2011. I define the set of dual advising firms based on behavior before Dodd-Frank. Because municipalities select into using “dual advisors,” my primary identification strategy accounts for selection by using within-issuer variation across both advisors and time, focusing on the set of competitively sold, general obligation, tax exempt bonds. The preferred specification indicates that borrowing costs decrease by 11.4 basis points (bps), or 5.3%, for dual advisor issues relative to independent advisor issues after regulation. The average treatment effect masks substantial heterogeneity: school districts for whom the dual advisor bid in most of their auctions before 2011 saw borrowing costs decline by 45 bps, while borrowers who get at least six bids on average have no change in borrowing costs after regulation. The effect is concentrated in the most opaque corners of the market where attention from other potential underwriters is relatively low.

Identification in the difference-in-differences research design assumes that outcomes for dual-advised and independently advised issues would have moved in parallel absent the regulatory intervention. As *prima facie* evidence in favor of the parallel trends assumption, I show that dual-advised and independently advised issues have similarly trending outcomes during the great recession in 2008–2009 and following years. Next, I show that the observed effects are not a function of changing composition of bonds and that the

⁴ The nomenclature used to describe the mechanics of the affiliated values auction here was introduced by [Hong and Shum \(2002\)](#), who also focused on a similar procurement auction where the low bid wins. For bidder i with signal x_i in an auction with n bidders, the general optimal bidding function can be written as

$$s_n(x_i) = \frac{s'_n(x_i)[1 - F_{-i,n}(x_i | x_i)]}{f_{-i,n}(x_i | x_i)} + v_n(x_i, x_i).$$

[Milgrom and Weber \(1982\)](#) showed that, under some general conditions for a low-bid auction, the first term is decreasing in n , which is the competitive effect, while the second term is increasing in n , which is the adverse selection effect.

yield decrease is concentrated entirely in new money issues and issues by opaque school districts where asymmetric information is most likely prevalent and scope for adverse selection is largest. Extended analysis in [Internet Appendix C](#) details an alternative identification strategy using inverse probability weights, robustness to measurement and specification decisions including advisor-by-issuer fixed effects, and a placebo test highlighting no effect of regulation on unaffected advisors in firms with investment banks.⁵

To shed light on the relative importance of potential mechanisms for explaining declining borrowing costs, I begin by measuring changes in auction participation and outcomes. This yields two pieces of evidence that the adverse selection effect is dominant. First, when the dual advisor is no longer allowed to bid, the total number of submitted bids increases by 0.4 bids relative to a median of 5 bids, indicating that the removed bids are more than fully replaced (i.e., the treated auctions have net entry after barring a historical participant). Second, the winning bid decreases (the winning bidder pays more) and this decrease in bids is observed across the whole distribution of bidders such that the average and median bid also decrease in the same magnitude as the winning bid, indicating that all remaining potential underwriters bid more aggressively, and it is not simply that the winning bid was luckier for the issuer. As additional evidence of the adverse selection effect, I document the presence of the *winner's curse* in 2008–2011 when nonadvisors win underwriting auctions while bidding against dual advisors. I find lower ex post gross spreads for nonadvisor underwriters who win in auctions against a dual advisor, consistent with nonadvisor underwriter wins being subject to adverse selection.

Next, I test whether the average structure of bonds is changing in the first stage in a manner consistent with changing advice. I find that bonds with potential dual advisors become more similar to bonds with independent advisors after 2011. Among dual-advised issues, the maturities for similar size issues become shorter, although this result is statistically weak, and the bonds are more likely to obtain a credit rating or use a credit enhancement.⁶ The newly rated bonds are similarly rated to the market as a whole, which suggests they were not eschewing ratings because of underlying creditworthiness differences. These patterns are consistent with conflicted dual advisors encouraging creditworthy issuers to avoid third-party certifications to increase private information that other underwriters would not have in order to increase adverse selection and their own profits. The increase in third-party certifications and

⁵ A similar update to MSRB Rule G-17, the *fair dealing rule* for underwriters of negotiated deals, was implemented in August 2012 and required that underwriters disclose their compensation structure and that they are not financial advisors. This regulation could impact the negotiated business of both dual and independent advisors and provides one reason for my focus on competitive sales. It is conceivable that the negotiated underwriting business of dual advisors is harmed by Rule G-17, which could conceivably lead to them trying to extract relatively more value from their competitive advisory business studied in this paper and could bias my estimates toward finding increases in costs for dual-advised auctions, which is the opposite of what I find. Rule G-17 is discussed more in [Internet Appendix B.1](#).

⁶ Credit enhancements in the market include bond insurance, letters of credit, and guarantees.

standardization manifests as more liquid trading on secondary markets.⁷ The decrease in borrowing costs is large when compared to the costs associated with obtaining third-party credit certifications, but Section 5 discusses how this comparison omits other potential costs.

The larger importance of the adverse selection effect relative to the competitive effect of removing an intermediary conflict of interest stands in contrast to much of the existing literature. Many papers find that the competitive effect—having more firms participate in the market—is more important to client outcomes than a potential conflict of interest. For example, Puri (1996), Gande, Puri, and Saunders (1999), and Drucker and Puri (2005) study the Glass-Steagall Act and its repeal and find that the impact of additional competition swamps the potential downside of allowing potentially conflicted agents from participating in the market. In many scenarios, even a single additional competitor is enough to dominate other market design characteristics (Bulow and Klemperer 1996). The paramount role of competition shows up repeatedly throughout the literature on financial intermediaries and the municipal bond market is no exception (Cestau 2019; Garrett et al. 2023). The academic focus on the importance of competition is consistent with the policy debate around Rule G-23, which centers on the marginal impact of losing the dual advisor's bid in the second stage (Bond Dealers of America 2019). The present study argues that competition, as measured by the participation in auctions, is a function of the dual advisor's conflict of interest because of the adverse selection effect. Firms are less willing to compete when information is endogenously withheld. While the competitive landscape of the industry is important, dual advisors were able to profit in the least competitive parts of the market by advising bonds that were issued in a more opaque manner.⁸

In the market for consumer financial advice, a similar conflict of interest arises from the dual function of some financial advisors who both give advice about what securities to buy and who also directly sell their own funds. The empirical literature has overwhelmingly shown that this conflict of interest drives financial advisors to give advice that lowers net yields for investors while enriching the firm through increased demand for its own investment products or through the ability to adjust their own portfolios (Foerster et al. 2017; Hackethal, Haliassos, and Jappelli 2012; Fecht, Hackethal, and Karabulut 2018; Hoechle et al. 2018; Mullainathan, Noeth, and Schoar 2012; Chalmers

⁷ I also test for changes in the quality of underwriting proxied by 30-day underpricing in Internet Appendix C.9 and find no statistically significant evidence of a deterioration in eventual underwriting quality although the difference-in-differences point estimates are mostly positive.

⁸ The headline estimates finding material impacts of a conflict of interest here are largely consistent with studies of other financial services where the intermediary controls information, such as credit rating agencies. Credit ratings agencies have been shown to use their informational advantage over clients to extract information rents (Jiang, Stanford, and Xie 2012; Griffin and Tang 2011). Similarly, integrated mortgage lenders are able to benefit from informational advantages relative to nonintegrated lenders to cream skim the least risky mortgages (Stroebel 2016).

and Reuter 2020; Boyson 2019; Bhattacharya, Illanes, and Padi 2019). In a related study on consumer financial advice, Egan, Matvos, and Seru (2019) show that advisory firms specialize in misconduct and profit from focusing on unsophisticated rich and less educated individuals. A similar mechanism shows up in this paper where certain types of advisors can deter competition by withholding information regarding small, opaque, and likely unsophisticated municipal issuers as well.⁹

This study also contributes to the literature that studies municipal borrowing costs and thus the cost of providing many public goods at a subnational level. Municipal borrowing costs vary widely across the United States with true interest costs ranging from 0.1% to over 8% in this sample from SDC Platinum (2016) for 2008 to 2015. Municipal borrowing costs are driven by factors including tax rates and exemptions (Poterba 1989; Fortune 1991; Ang, Bhansali, and Xing 2010; Cestau, Green, and Schürhoff 2013; Liu and Denison 2014; Garrett et al. 2023), market segmentation and illiquidity (Schultz 2012; Schwert 2017; Cestau et al. 2019; Babina et al. 2021), market structure faced by the issue (Cestau et al. 2021; Cestau 2019; Ivanov and Zimmerman 2019), environmental risk (Goldsmith-Pinkham et al. 2023), racial bias (Dougal et al. 2019), and many other local characteristics and decisions (Poterba and Rueben 2001; Cornaggia, Hund, and Nguyen 2019; Gao, Lee, and Murphy 2020). The choice of financial intermediaries affects borrowing costs and these agents are often chosen through political connections (Simonsen and Hill 1998; Butler, Fauver, and Mortal 2009) or by geographic proximity (Butler 2008). Advisors have wide breadth to affect borrowing costs by changing the structure of municipal bonds, by their differing abilities to find underwriters, and by their reputations for working with municipalities to create successful issues (Clarke 1997; Liu 2015; Moldogaziev and Luby 2016; Bergstresser and Luby 2018; Daniels et al. 2018). This study adds a novel finding to the public finance literature that conflicting financial incentives of advisors also negatively affect outcomes for municipal bonds by endogenously increasing asymmetric information and thus adverse selection, especially in low-participation corners of the market.

1. Financial Advisors and MSRB Rule G-23

Municipalities issue over \$400 billion dollars of bonds each year to finance investment in public goods, such as water treatment plants, roads, and schools. Interest payments on these bonds make up one of the largest public expenditures for state and local governments. Some of the investments financed by municipal debt raise property values and tend to be associated with other

⁹ Increased information provision lowering borrowing costs is also consistent with other research on how information affects the municipal bond market, such as Baber and Gore (2008), and is related to the larger literature on information and the cost of capital (Duarte et al. 2008).

positive economic outcomes (Dagostino 2019; Adelino, Cunha, and Ferreira 2017; Cellini, Ferreira, and Rothstein 2010). However, several frictions in the market increase borrowing costs for public entities. First, the market to hire underwriters is not perfectly competitive (Garrett et al. 2023; Cestau et al. 2019). Second, municipalities often lack the financial sophistication to navigate the issuance process on their own because they issue debt infrequently in a market segmented into narrow markets by local tax exemptions and regulation (U.S. Securities and Exchange Commission 2012; Schultz 2012; Bergstresser and Luby 2018; Babina et al. 2021).

A municipality that wishes to invest in a new public project can be imperfectly informed about the market for their debt. The municipality may hire a municipal financial advisor (henceforth, “advisor”) to develop a financial plan to raise funds through a bond offering or other debt instrument and how they will pay for it. After a municipality decides to issue bonds, the advisor helps structure the debt, create public disclosure documents, solicits a credit rating or credit enhancement, and makes a plan to find an underwriter who will certify the debt and sell the bonds to investors on a secondary market. I focus on competitive sales where an underwriter is picked by submitting the lowest yield to maturity in a first-price, sealed bid auction. The underwriter is responsible for certifying the quality of the bond issue to the market and for selling the bonds to investors or other broker-dealers.

The process of issuing municipal debt in the United States is regulated by the Municipal Securities Rulemaking Board (MSRB), which develops rules regarding how advisors, underwriters, and issuers are allowed to interact. Since its founding by Congressional mandate in 1975, the MSRB has been charged with protecting investors and the public interest from financial malfeasance by municipalities and financial institutions. The MSRB created Rule G-23 in 1978, which was primarily geared toward disclosure of roles. The original rule required that advisors would have to disclose their intent or ability to underwrite to municipalities that they were advising. Potential dual advisor underwriters would need to get permission from the issuer to potentially fulfill a dual role. Historically, it was common for advisors to act as both advisor and underwriter for a given bond issue. Dual advisors underwrote about 15% of the competitive issues they advised before 2011.

The underwriter and municipality have different and potentially opposing goals when issuing debt: a municipality that wants to minimize financing costs will want a low interest cost but the underwriter wants to maximize profits, which increase with interest costs (holding investor demand and other factors constant). Even if taking a specific action lowers the eventual price investors will pay for a bond, as long as the price that the underwriter pays to the municipality decreases more, then this would be a profit increasing action for the underwriter in their role as an advisor. Schapiro (2010) summarizes the SEC’s opinion on the conflict of interest:

“Financial Advisers should be prohibited from resigning as financial advisor to an issuer, and then underwriting that issuer’s bonds, as they are currently allowed to do under MSRB Rule G23. Right now, a financial professional advising a municipality can guide the municipality towards securities tailored to his firm’s advantage, then resign and act as underwriter. This is a classic example of conflict of interest.”

In response to the financial crisis of 2007–2008, Congress passed the Dodd-Frank Wall Street Reform and Consumer Protection Act in July of 2010, which added a new charge to the MSRB’s function in Subchapter H of Title IX. In addition to protecting investors and the public interest, the MSRB also became responsible for protecting states and other municipal entities who sell debt through financial intermediaries. To do so, the MSRB was given the power to regulate municipal financial advisors separately from underwriters and broker-dealers.

The first new regulation from the MSRB after Dodd-Frank was a restatement of Rule G-23, which shifted from focusing on disclosure to a prohibition of behavior. On August 17, 2010, the MSRB posted a request for comments regarding how Rule G-23 could be changed to better reflect the new functions under Dodd-Frank. By February 2011, the MSRB filed the final language with the SEC and, on May 31, 2011, the SEC officially approved the final changes to Rule G-23 that prohibited all municipal financial advisors from underwriting any debt about which they offered advice. The regulation came into effect 6 months later on November 27, 2011 (SEC 2011). Bergstresser and Luby (2018) review the other changes facing advisors after Dodd-Frank including a registration rule in 2014, a fiduciary rule after 2015, and additional licensing requirements in 2017 that affect all municipal financial advisors.¹⁰

2. Municipal Bond Market Data and Outcomes

Analysis data come from the SDC Platinum Global Public Finance database (SDC Platinum 2016). SDC lists all municipal bond issues with a full description of the characteristics of the issue. The data include the sale date, the principal size, an account of individual bonds in each issue and associated coupons, maturity dates, CUSIP codes, and many other bond characteristics. See Internet Appendix A for definitions of all variables used from SDC. These data provide the basic information to estimate a pricing model of municipal bond issues in the primary market and they include the names of the financial advisor and underwriter.

¹⁰ While there is potential for dual advisors to change their behavior during the period between Dodd-Frank and the official start of the rule, the lack of any pre-trends in yields between 2008 and 2011 in the empirical analysis suggests that there was not an important response to the threat of regulation before it happened.

The SDC data include listings of bonds that are sold via direct negotiation with an underwriter and those sold competitively at auction. I focus on competitive sales for three primary reasons. First, advisors are almost always present in competitive sales while their potential actions as underwriters are differently regulated in the negotiated market. Second, most bond sales that take place competitively are statutorily required to do so while negotiated sales almost always represent an endogenous choice on the part of the issuer (Cestau et al. 2021). Third, the use of advisors is undergoing a dramatic increase in the negotiated market where they offer a different set of services than what issuers require for competitive sales and where regulation binds differently. Competitive sales also have the added benefit of being able to directly observe competition related mechanisms since bids become public information after the auction.¹¹

The second data set provides networks of municipal advisors, investment banks, and financial holding companies that provide municipal underwriting services developed by Bergstresser and Luby (2018). This list identifies municipal advisors in the SDC data that are associated with an underwriting entity. I call such advisors who also offer broker-dealer services through another arm of the same business “dual advisors” if they are ever observed advising and underwriting the same competitive deal between 2008 and 2010. The behavior of these dual advisors is controlled by the reformed version of MSRB Rule G-23. Advisors that are not associated with an investment bank cannot act as a broker-dealer—whether or not such behavior is forbidden by Rule G-23—so their behavior is not directly affected by the 2011 reform. I refer to those financial advisors as “independent” or “dedicated” advisors.¹² Four investment banks have advising lines of business but never submit bids on issues on which they offered advice in the sample. The advising branches of these firms are used in a placebo test that shows no treatment effect for advisors who never engaged in dual advisor behavior.¹³ The market for municipal financial advice is made up of many types of firms, from companies advising over 500 competitive issues per year down to advisors that advise less than one issue per year on average. The 15 largest dual advisors from 2008 to 2011 are shown in Table A.1. The largest of these advisors is FirstSouthwest. Before the 2011 reform, FirstSouthwest advised over 300 issues per year, while smaller advisors like D. A. Davidson & Company and GMS Group LLC both advised fewer than 10 issues each year.

Next, I match the SDC and advisor firm structure data with first-price, sealed bid auction results from The Bond Buyer (2016). The Bond Buyer, one of the primary trade publications for municipal bond underwriting and trading,

¹¹ See Internet Appendix B.1 for more discussion of the focus on competitive sales and the history of advisors in negotiated sales.

¹² The geographic distribution of dual advisors and borrowing behavior is discussed in Internet Appendix B.

¹³ See Internet Appendix C.5 for more information and results of the placebo test.

is a regular source of information regarding the primary market for municipal bonds. Starting in 2008, they began publishing the results of recent competitive auctions for underwriting privileges with bids for all participants in addition to advance notices of upcoming auctions. I match the primary market auction results from The Bond Buyer to SDC based on the issuer name and state, issue size, issue date, and underwriter to establish a list of 41,182 competitive municipal bond auctions from February 21, 2008, through December 31, 2015. I further restrict the sample to those issues with principal greater than \$1 million, those funded by general obligation,¹⁴ issues that employ an advisor,¹⁵ and those qualifying for the federal tax exemption to find a final set of 20,051 auctions made up of 286,042 individual bonds with principal value of \$349 billion. Dual advisors are employed in 5,735 of the auctions. Municipal Bonds are generally issued in series with 2–20 bonds in a single auction. The unit of observation for the main analysis is the auction level.

The median auction in the sample receives bids from five potential underwriters and the modal auction receives 4 bids, but this masks substantial heterogeneity. Thirteen percent of the sample receives only 1 or 2 bids, while 11% of the sample receives 9 or more bids. The distributions of the number of bids submitted in these auctions are shown in Figure 1, separated by advisor type and whether the auction happened before or after November 27, 2011. The changes between the pre- and post-Rule G-23 bid distributions for each advisor type are shown in panel C, which provides a preliminary difference-in-differences estimator for the change in very low participation issues. Before regulation, 30% of dual-advised issues received three or fewer bids. After regulation, only 22% of dual-advised bonds received three or fewer bids, which is an 8-percentage-point decline. Over the same period, all other advisor issues only decreased the share of auctions with three or fewer bidders from 26% to 24%. The change in auction participation over time is revisited with controls in a formal difference-in-differences analysis in Section 3.2. The auction data also shed light on how frequently dual advisors bid on the issues they advise, even if they don't win, which I display in Table A.1. FirstSouthwest bid in 49% of issues they advised, very close to the average, while another major dual advisor UniBank Fiscal Advisory Services only bid in 15.5% of the issues that they advised. This variation in bidding likelihood provides additional variation for later empirical tests.

Secondary market transaction data are gathered from the [Municipal Securities Rulemaking Board \(2019\)](#). The MSRB Historical Transactions data report every trade involving a registered broker-dealer—purchases and sales,

¹⁴ Bonds backed by “general obligation”—referred to as general obligation bonds—are usually backed by the full faith and credit of the issuing municipality, meaning that unrelated tax streams may be used to pay off the debt and any revenues from the project are not specifically earmarked for debt repayment.

¹⁵ Almost all competitive sales employ a financial advisor. Restricting on having an advisor eliminates some very short-term issues as well as some refunding issues. This also eliminates debt that is likely very simple in terms of structure, which is not a good comparison group to bonds with advisors.

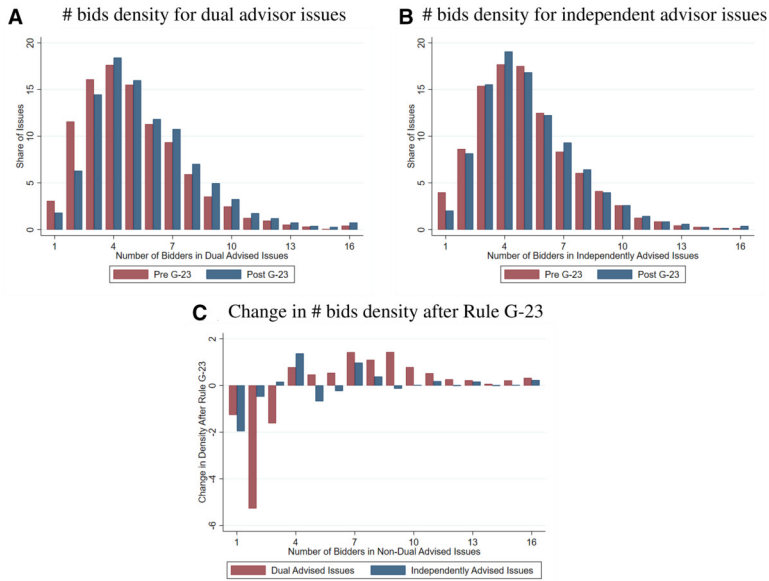


Figure 1
Distribution of ex post submitted bids by advisor type and period

Histograms illustrate the number of bids submitted for issues with different types of advisors and in different time periods. Panel A shows the observed distribution of submitted bids for issues with dual advisors. The maroon bars represent the period before November 27, 2011, and the navy bars represent the period after once Rule G-23 is implemented. Panel B shows the same distributions for independently advised issues before and after Rule G-23. The difference between the pre- and postperiod for each sample is displayed in panel C. For dual-advised issues, the share of auctions with 1–3 bidders declines by 8 percentage points, which matches the increase in the share of auctions with 4 or more bidders. For independently advised issues, the analogous decline in low participation auctions is only 2 percentage points. Author’s calculations using data from the [The Bond Buyer \(2016\)](#), [SDC Platinum \(2016\)](#), and [Bergstresser and Luby \(2018\)](#).

separately—in the over-the-counter municipal bond market. I match the MSRB prices to their primary market issues by matching CUSIPs at the bond level to the SDC data. These data allow me to measure differences in secondary market outcomes that arise because of regulation of the primary market. The data report trade date and time, trade price, trade yield, trade size, and whether the counter party was a final investor or a broker-dealer.

2.1 Selecting a dual advisor for a particular competitive sale

To test whether selection is likely to bias the estimates of a difference-in-differences research design, I examine what sort of issuers choose to hire dual advisors and what sort of issues dual advisors may specialize in. I begin by estimating a linear probability regression describing the choice of dual advisor for each bond issue,

$$\mathbb{1}\{dual_{ijt} = 1\} = \eta_t + X_{ijt}\beta + \varepsilon_{ijt}, \quad (1)$$

where subscript i indicates the issue, j indicates the issuer, and t indicates the time. X_{ijt} includes variables controlling for size, years to maturity, refunding

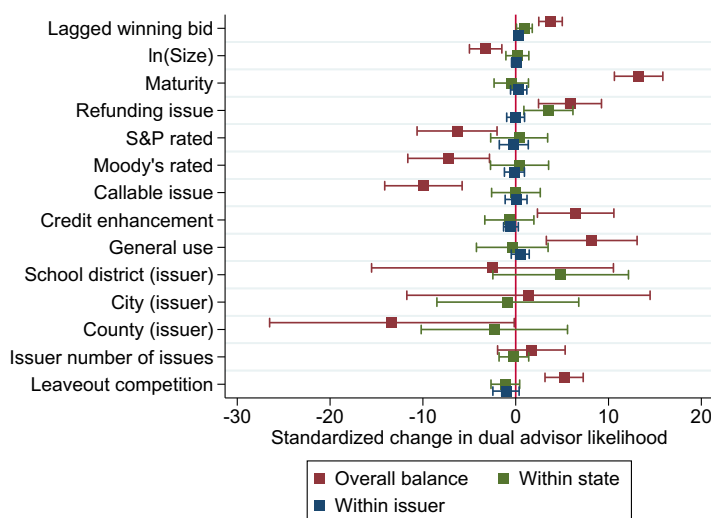


Figure 2

Linear probability estimates explaining dual advisor choice, 2008–2011

Boxplots depict the point estimates and 95% confidence intervals from regressions of choice of dual advisor on issue and issuer characteristics for competitive auctions. Characteristics with continuous measurements are normalized to standard deviations. Numerical estimates and additional specifications are available in Table A.3. Overall Balance, corresponding to column 1 of Table A.3, shows that smaller issues, issues with longer maturities, refunding issues, unrated issues, and issuers with more participation in auctions are more likely to have dual advisors. Further, school districts, towns, cities, and county issuers are less likely to use dual advisors than states, state agencies, and special districts (the omitted category). Within State balance, corresponding to column 3 of Table A.3, replicates this regression including state fixed effects and shows that most of the variation in observable characteristics for issues with dual advisors is explained by geographic variation, but refunding issues are less likely to choose dual advisors while bonds for general public improvement (General Use) are less likely to employ dual advisors. Within Issuer balance, corresponding to column 5 of Table A.3, shows that the bonds that use dual advisors are not observably different than bonds with independent advisors after accounting for average issuer characteristics. Standard errors are clustered at the issuer level, and 95% confidence intervals are included. Author's calculations using data from SDC Platinum (2016), The Bond Buyer (2016), and Bergstresser and Luby (2018).

status, credit ratings, use of funds, type of issuer, frequency of issuer borrowing, and average auction participation (number of bidders) for other issues for the same issuer.

I estimate this regression with three different sets of fixed effects to highlight the size and direction of selection with different sources of variation. Figure 2 shows the estimated coefficients for all three specifications using issues from 2008 to November 2011, before any reforms from Dodd-Frank take effect. Point estimates and standard errors are included in Table A.3. The results from the regression without any geographic controls suggests that municipalities select advisors based on characteristics of issues. The variation in issue characteristics is mostly explained by spatial variation across states in the average type of issue. With state fixed effects, the effects of most controls disappear except for refunding issues, which are still more likely to choose dual advisors. Further, the issues from issuers with more auction participation

on average are less likely to choose dual advisors after controlling for state differences, reversing the overall pattern.

Figure 2 also shows estimates of Equation (1) using issuer fixed effects. In this specification, variation in selection comes from issuers who issue multiple times choosing different advisors for sequential issues. After controlling for issuer fixed effects, all of the observable differences between dual advisor and independent advisor issues go away. Using issuer fixed effects, the identifying variation for the analysis in the following sections comes from two sources: issuers who never change their advisor for all observed issues and issuers who switch advisors in both pre- and postreform periods. This switching behavior is not related to Dodd-Frank or the new regulation in 2011, and it is documented more fully in [Internet Appendix B.3](#).

3. Borrowing Costs after Rule G-23

Borrowing costs are the outcome of interest for municipal borrowers where we may expect the effects of a conflict of interest to manifest. A conflicted dual advisor who also intends to underwrite a given issue could potentially profit by increasing the yield in the primary market as long as the yield at which they place bonds with investors increases by less. If borrowing costs in the primary market auctions increase after regulation, it suggests that the competition provided by the dual advisor in the second stage was the most important factor for client outcomes. If borrowing costs decrease, it suggests the adverse selection effect is most important, and motivates further questions about why adverse selection is important in these transactions.

To measure what happens to issuer borrowing costs after the reform of MSRB Rule G-23, I begin by estimating a standard difference-in-differences regression ([Bertrand, Duflo, and Mullainathan 2004](#)):

$$b_{1ijt} = \alpha \text{dual}_i + \beta \text{dual}_i \times \text{post}_t + X_{ijt} \Gamma + \lambda_j + \lambda_t + \varepsilon_{ijt}, \quad (2)$$

where b_1 is the winning bid for an auction, which is measured as the true interest cost (TIC) of the bond series. The results are robust to defining the outcome as the yield-to-worst instead of TIC.¹⁶ Subscript i indicates the bond series, subscript j denotes the issuer, while subscript t indicates the time. X_{ijt} includes control variables that influence the value of the bond package. The measures of TIC, which are a specific calculation of yield-to-maturity defined by the MSRB, primarily come from [The Bond Buyer \(2016\)](#) when available and are otherwise imputed from [SDC Platinum \(2016\)](#).

¹⁶ TIC is a standard measure of interest cost in the municipal finance literature, but it is not a complete measure of financing costs because it ignores early repayment risk ([Luby and Orr 2019](#)). [Internet Appendix C.3](#) details other calculations of interest costs that take prepayment risk into account and shows that results are robust to the measurement of interest costs.

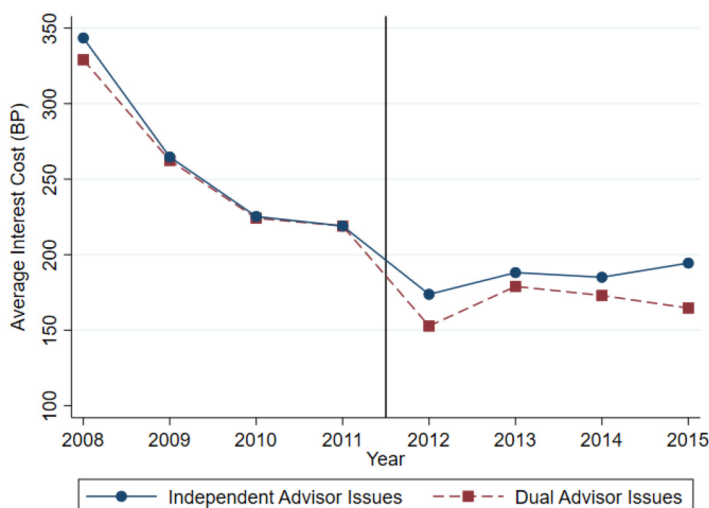


Figure 3

Average borrowing costs by advisor type, no controls

This figure reports the annual average borrowing costs for dual advisor issues and independent advisors. Borrowing costs are residualized by removing issuer fixed effects. The level for dual advisor issues is normalized such that the mean borrowing cost is equal to the independent advisor level in 2011. See Section 3 for more information and discussion. Author's calculations using data from [SDC Platinum \(2016\)](#), [The Bond Buyer \(2016\)](#), and [Bergstresser and Luby \(2018\)](#).

The baseline controls include issuer fixed effects, year fixed effects, and controls for state economic conditions and fiscal policies.¹⁷ Additional specifications add controls for market conditions, fixed effects for refund status, fixed effects for callable bonds, a control for natural logarithm of size that is allowed to vary by year, flexible time trends for different types of municipal entities, and fixed effects for three bond maturity bins. post_t is an indicator that is equal to one if an issue takes place after November 27, 2011, and zero otherwise. dual_i is an indicator variable for issues that are advised by a dual advisor defined by the advisor's behavior before 2011. β is the change in interest costs for issues with dual advisors after the reform of Rule G-23, which is the treatment effect of interest. All regressions are estimated with standard errors clustered at the issuer level.

The unconditional average borrowing costs for dual advisor issues and independent advisor issues normalized to match in 2011 are shown in Figure 3. This figure highlights that the parallel trends assumption appears to hold in the data with no controls in the X_{ijt} matrix. Further, this figure shows how aggregate trends in municipal borrowing costs are decreasing through

¹⁷ The state economic condition and policy controls include income tax rates from which municipal bonds are exempt, logarithm of state gross domestic product (GDP), logarithm of intergovernmental transfers, and unemployment rates averaged to the year level.

Table 1
Difference-in-differences model explaining primary market interest cost

	(1)	(2)	(3)	(4)	(5)
Dual advisor	6.085 (6.667) .361	5.248 (6.637) .429	5.314 (6.621) .422	7.308 (4.845) .132	3.764 (3.804) .323
Dual advisor X Post	-12.500 (3.453) .000	-11.778 (3.425) .001	-11.296 (3.424) .001	-13.149 (2.720) .000	-11.382 (1.970) .000
Observations	20,051	20,051	20,051	20,051	20,051
Mean interest cost (BP)	216.759	216.759	216.759	216.759	216.759
Year and issuer FE	Y	Y	Y	Y	Y
State economic and policy controls	Y	Y	Y	Y	Y
Market climate controls		Y	Y	Y	Y
Issuer type-by-year FE			Y	Y	Y
Size, refund, and call controls				Y	Y
Maturity terciles					Y

This table shows the estimates from regressions of primary market issue outcomes on type of advisor before and after the MSRB Rule G-23 reform for all competitive, general obligation, and tax-exempt issues matched between the SDC Platinum database and Bond Buyer that employ financial advisor. The dependent variable is the interest cost in basis points. All specifications control for year fixed effects issuer fixed effects, and state economic and policy controls. Column 2 adds controls for market conditions with SIFMA yields and 1- and 10-year swap spreads. Column 3 adds flexible trends for different types of issuers. The specification in column 4 adds controls for bond characteristics intrinsic to the project including size, refund status, and callability, while column 5, the preferred specification, adds fixed effect for years to maturity. See Section 3 for more information and discussion. Standard errors clustered at the issuer level are shown in parentheses with *p*-values below. Author's calculations using data from [SDC Platinum \(2016\)](#), [The Bond Buyer \(2016\)](#), and [Bergstresser and Luby \(2018\)](#).

the sample, but the aggregate trends are affecting dual advisor issues and independent issues in the same manner in the preperiod, including during the period when the Rule G-23 regulation was first being discussed.

Estimates of α and β from Equation (2) are shown in Table 1. The question this regression answers is whether the policy-relevant outcome—borrowing costs—changes differently for dual-advised issues than issues with independent advisors before and after the change to MSRB Rule G-23. The first column shows the estimates with issuer fixed effects, year fixed effects, and controls for state economic conditions. In this specification, interest costs fall by 12.5 bps for dual-advised issues after the reform of Rule G-23, significant at the 0.001% level.¹⁸ A causal interpretation of this regression is that regulating the conflict of interest from dual advisors by prohibiting underwriting lowers borrowing costs for treated issues by 12.5 bps, or about 5.8%, relative to the mean of 216 bps, relative to issues with independent advisors.

The identifying assumption behind this causal interpretation is that variance-weighted interest costs would change in parallel for bonds with dual advisors and independent advisors within the same issuer. I estimate four more specifications with increasing controls to capture other market changes and bond characteristics that could conceivably drive this result.

¹⁸ Issues that use a dual advisor have interest costs that are 6.085-bps higher on average in the preperiod. This is not statistically significant.

The second column of Table 1 adds three flexible controls for market fluctuations on the day that the underwriter auction is held. First, [SIFMA \(2019\)](#) reports a swap rate index for AAA-rated municipal variable rate debt obligations (VRDOs) that captures fluctuations specific to the municipal market. Second, I add controls for 1- and 10-year swap rates for the 3-month LIBOR reported by [Board of Governors of the Federal Reserve System \(2018\)](#). These controls adjust for different preferences for market timing that interact with secular market trends and risk, but the addition of market risk controls has no effect on the estimated coefficients. Flexible time trends for different types of issuers are added in column 3. The functional form is a fixed effect for each issuer type—school districts, counties and special districts, states and state agencies, townships, cities, and other issuers—interacted with year fixed effects. This captures how the secular trends affect different types of borrowers since advisors may specialize in funding certain types of projects that change around the same time as the Rule G-23 reform. The inclusion of flexible time trends does not have a material effect on the estimates.

Column 4 of Table 1 adds several controls for issue specific characteristics that are generally driven by the nature of the project financed by the bond instead of by advice from the advisor. The controls include a fixed effect for the refunding status of the bond,¹⁹ a control for the natural logarithm of the principal value of the bond, and indicators for the callability of the bond. Controlling for the bond characteristics that the advisor does not directly influence does not change the estimated coefficients in a material way. Finally, I add fixed effects for the final maturity of each issue cut into three bins in column 5. The measurement in relatively coarse bins in this case is because advisors may be able to influence the time to maturity as I will show in Section 4.1, which makes precise fixed effects for time to maturity endogenous controls although their inclusion does not have a material impact on the estimates. I also do not include controls for credit rating or credit enhancements in the baseline specification because these are margins over which dual advisors change behavior as shown in Section 4.1. The presence of idiosyncratic shocks to credit worthiness is only a threat to identification insofar as the shocks are correlated with the choice of using a dual advisor, which would show up in the pretrends during 2008 and 2009, when markets were much more volatile than in later years in the sample. To further rule out confounding effects of correlated shocks to underlying issuer credit worthiness and propensity to employ a dual advisor, I include several specifications that restrict the set of issuers to those whose issuing behavior and other observable characteristics are unchanging during the sample in [Internet Appendix C.4](#).²⁰

¹⁹ The primary categories for this variable are refunding, advanced refunding, and new money.

²⁰ Additional specifications including endogenous controls that were included in early versions of this paper are shown and discussed in [Internet Appendix C.1](#) with coefficients varying between -8.2 and -10.7 bps.

The estimates from the fully saturated specification in column 5 indicate that borrowing costs for dual advisor issues fall by 11.4 bps (p -value $< .001$), which is 5.3% of the average yield in the sample, relative to the interest costs for comparable issues with independent advisors. This result is stable across specifications, suggesting that observable changes in bond characteristics and other market trends are not driving results. In [Internet Appendix C.4](#), I show that all participating underwriters, including the losing underwriters, bid more aggressively for dual advisor bonds after the regulation such that the average and median bids also decrease in the same magnitude.

The event study with annual coefficients associated with the specification in column 5 of Table 1 is shown in Figure 4, normalized such that the difference in borrowing costs in 2011 is equal to zero. Each annual coefficient is the interaction of dual_{*i*} with a year indicator.²¹ In the preperiod, dual advisor issues were more expensive than issues with independent advisors, and there is a drop of 10 bps immediately in 2012. The following years in the postreform period show borrowing costs continue decreasing for dual advisor issues with estimates ranging from −10 bps in 2013 to −22 bps in 2015. Figure 4 also serves as an informal check to the assumption of parallel trends. A visual examination of the graph reveals no observable pre-trends in the treatment effect after hiring a dual advisor instead of an independent advisor since the preperiod estimates are all within 2 bps of each other. Further, the years 2008 to 2010 were very volatile for municipal bonds, and the lack of differential trends in those years highlights that general volatility does not affect dual-advised and independently advised issues differently.

3.1 Composition and borrowing cost robustness

A potential concern with the analysis so far is that the set of bonds being taken to market could be responding to the regulation and the observed interest cost effect could reflect a change in market composition not captured by the rich set of controls. Several recent papers have shown that the number and amount of bonds that municipal entities issue increase as yields decrease ([Dagostino 2019](#); [Yi 2020](#); [Haughwout, Hyman, and Shachar 2021](#)). Notably, [Adelino, Cunha, and Ferreira \(2017\)](#) show that municipalities borrow more in response to credit upgrades. In the case of MSRB Rule G-23, if borrowing costs actually increase for affected municipalities so much that marginal borrowers no longer issue, they may drop out of the market entirely. If marginal borrowers have a higher interest cost to start with, their exit from the market could be conflated with a decrease in borrowing costs. This potential mechanism can be directly tested by measuring the extensive and intensive quantity responses to the regulation. First, I test whether issuers who issued bonds with a dual advisor pre-2011 are more or less likely to issue after the regulation than other

²¹ Dates after November 26, 2011, are combined with 2012, since the updated MSRB Rule G-23 came into effect on November 27, 2011.

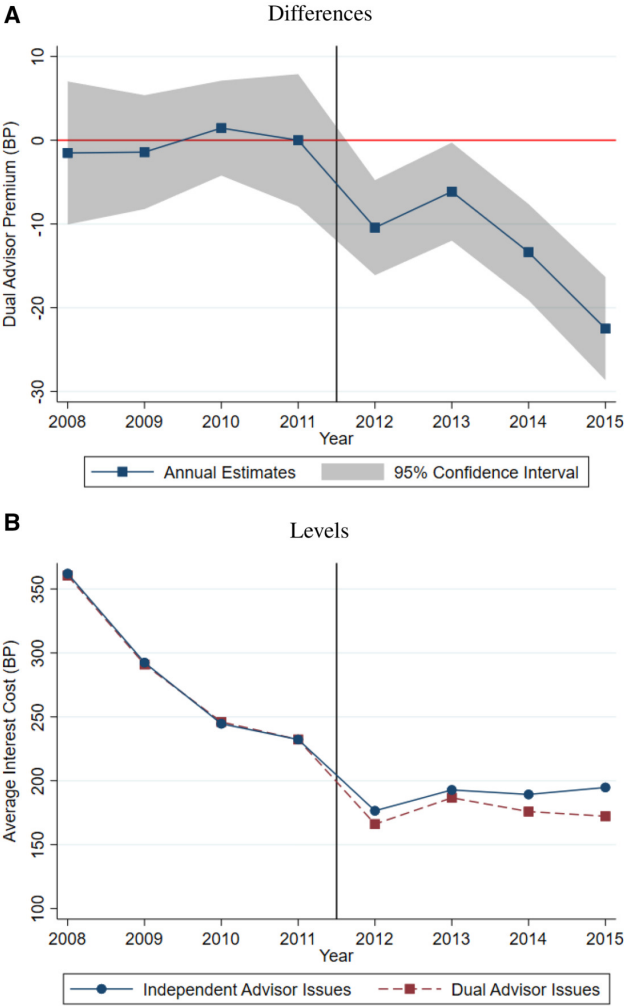


Figure 4
Treatment effect of dual advisor on winning bid (basis points)

This figure reports the annual effects of having a dual advisor on borrowing costs in basis points as estimated by Equation (2). The levels in the graphs are scaled such that the mean effect in 2011 is zero. The specification reported in this figure corresponds to column 5 in Table 1 using all controls. See Section 3 for more information and discussion. Standard errors are clustered at the issuer level, and 95% confidence intervals are included. Author's calculations using data from [SDC Platinum \(2016\)](#), [The Bond Buyer \(2016\)](#), and [Bergstresser and Luby \(2018\)](#).

Table 2
Likelihood of issuance 2012–2015 as a function of preperiod dual advisor use

	(1)	(2)
Share dual advisor	0.030 (0.060) .623	0.031 (0.057) .597
Observations	4,528	4,524
Issuer type FE		Y

Author’s calculations using data from [SDC Platinum \(2016\)](#), [The Bond Buyer \(2016\)](#), and [Bergstresser and Luby \(2018\)](#). This table shows the estimates of a linear probability model describing the likelihood of issuers observed in the preperiod of issuing at least one bond in the postperiod. The sample is restricted to the 4,528 issuers who are observed in the preperiod. The independent variable is the share of bond issues advised by a dual advisor by each issuer before 2011. The second column adds an issuer type fixed effect to capture trends in the types of municipal entities that are seeking financing. Both specifications show that increasing the share of issues with a dual advisor in the preperiod increases the likelihood of issues after regulation, although the effect is insignificant. This rules out the worry that issuers are losing access to the market as their advisors are regulated. See Section 3.3 for more information and discussion. Robust standard errors are shown in parentheses with *p*-values below.

Table 3
Difference-in-differences model explaining count and volume of bonds issued

	(1)	(2)	(3)	(4)
Share dual advisor X Post	0.079 (0.100) .433	0.090 (0.100) .368	−0.000 (0.041) .997	0.004 (0.041) .928
Observations	5,762	5,758	5,762	5,758
Outcome	Count	Count	Volume	Volume
Period and issuer FE	Y	Y	Y	Y
Period-by-issuer-type FE		Y		Y

This table shows the estimates of a linear regression model describing the number and par value of bonds issued by each issuer as a function of preperiod dual advisor use. The sample is restricted to the 2,881 issuers who issue bonds in both periods. The independent variable is the share of bond issues advised by a dual advisor by each issuer before 2011. The independent variable is the number of issues in the first two columns and natural logarithm of aggregate par value in the third and fourth columns. The second and fourth columns add an issuer type-by-period fixed effect to capture trends in the types of municipal entities that are seeking financing. All specifications show that increasing the share of issues with a dual advisor in the preperiod is not associated with a change in the equilibrium quantity of bonds issued. This rules out the worry that issuers are losing access to the market as their advisors are regulated. See Section 3.3 for more information and discussion. Standard errors clustered at the issuer level are shown in parentheses with *p*-values below. Author’s calculations using data from [SDC Platinum \(2016\)](#), [The Bond Buyer \(2016\)](#), and [Bergstresser and Luby \(2018\)](#).

municipalities by using a linear probability model. In all specifications, issuers with dual advisors are more likely to issue after regulation, not less likely, which would bias the interest cost results toward zero although the effect is always statistically insignificant. Second, I change the dependent variable in Equation (2) to be either the number or aggregate par value of issues before and after regulation for a given issuer and aggregate bonds to the issuer level. I fail to reject the null hypothesis that the quantity of issues is unaffected by the use of dual advisors around the change to MSRB Rule G-23. The results of these regressions are shown in Tables 2 and 3.

A related general equilibrium concern is that the underwriting arms of dual advisors may change their behavior after Rule G-23. These underwriters

used to regularly underwrite issues that their firm advised, and now they have a requirement to only underwrite issues that their firm does not advise. This equilibrium response will create extra underwriting supply for non-dual-advised issues and creates a potential SUTVA violation: control units may have some partial spillover from the treatment issues as the removed bid doesn't necessarily disappear but moves from treatment to control. This bias would push me toward finding (1) increases in auction participation and (2) decreases in yields in independently advised (control) issues relative to dual-advised (treatment) issues. Both of these potential biases are the opposite of the empirical results, suggesting that my estimates could be somewhat attenuated toward zero relative to the true data generating process.

The estimates from the saturated specification in column 5 of Table 1 are robust across other specifications. In [Internet Appendix C.2](#), I show that these results are robust to a different identification strategy using [Hirano, Imbens, and Ridder \(2003\)](#) weights. In the robustness check, I estimate probabilities of using a dual advisor in the preperiod in a first stage. I calculate a counterfactual probability of choosing a dual advisor for each bond based on observables. The inverse of the probability of choosing the observed advisor type is used as a weight in a weighted least squares regression, which finds that bonds with dual advisors see interest costs fall by 7.1 bps after Rule G-23 was updated. [Internet Appendix C.4](#) discusses additional specifications with advisor and advisor-by-issuer fixed effects, setting the choice of dual advisor equal to the preperiod average share, defining dual advisors by the share of auctions on which each dual advisor bids, and with a restricted sample to consistent issuers. Finally, [Internet Appendix C.5](#) presents a placebo test using investment banks that sell both financial advice and underwriting services but never on the same bond issue before 2011. The placebo advisors have no change in borrowing cost after the regulation.

3.2 Auction participation

The major concern of attempting to limit a conflict of interest by removing the conflicted advisor from the second stage auction is that removing the dual advisor from the set of potential underwriters will decrease participation in the underwriter market, which would increase interest costs for bonds with a dual advisor. The analysis on borrowing costs above shows that, even if participation is decreasing, the policy-relevant outcome is improving on average. However, if auction participation is decreasing, it may indicate that the policy of prohibiting advisors from underwriting negatively affects client outcomes as well through the competitive channel. Below, I will use the difference-in-difference regression from Equation (2) to measure the change in auction participation, which I measure as the number of bids submitted in the underwriter auction, to test what happens to competitive forces in the underwriter market. This formalizes the differences in participation by advisor type shown in panel C of Figure 1.

I start by using the same controls as above in a regression explaining the number of bids submitted because factors that affect borrowing costs are also going to affect profitability for underwriters and marketability of debt to final investors. In addition to these controls, I also include controls for the number of potential underwriters who are likely to consider submitting bids in a given auction. I follow the methods of [Roberts and Sweeting \(2016\)](#) and [Athey, Levin, and Seira \(2011\)](#) to identify a set of unobservable potential bidders for each auction.²² I define potential underwriters as all underwriters who submit bids for bond issues that are in the same state-issuer type bin (e.g., school districts in North Carolina) as well as in the same principal size quintile in the 365 days leading up to each auction, plus the actual underwriters who submit bids. The advisor is removed from the set of potential bidders in the postperiod if they have recently bid in similar auctions since they are legally not allowed to underwrite bonds that they advise. Controlling for potential bidders in this way forces me to drop 2008 from the underwriter competition regressions since I do not observe 365 days of bidding behavior before those auctions in order to construct the potential bidder measure.

The estimates from regressions of the number of bidders on dual advisor and dual advisor interacted with the postreform dummy variable, conditional on number of potential bidders-by-year, are shown in Table 4. These regressions show that dual advisor issues had less competition before MSRB Rule G-23. Point estimates of the preperiod effect of hiring a dual advisor fall between -0.612 and -0.647 and are significant at the 0.001% level in all specifications. After MSRB Rule G-23, total auction participation increases by 0.423–0.462 underwriters, all significant at the 0.001% level. In the preferred specification in column 5, the joint effect of hiring a dual advisor in the postreform period is not statistically distinguishable from zero (-0.189 bidders with a p -value of .243). This indicates that dual-advisor issues had less competition than comparable issues with independent advisors before regulation, but this difference goes away after the dual advisor is removed from bidding. These results are also consistent with the unconditional difference-in-differences presented in panel C of Figure 1 that show the number of auction participants increasing for dual advisor issues after Rule G-23 and converging to roughly the same number of bidders as independently advised issues over time.

The median bond issue has five underwriters submit bids, so an increase of 0.42 underwriters represents a 8.4% increase in the number of underwriters competing for business. Focusing on nonadvisor underwriters, I repeat the analysis above using the number of nonadvisor bids as the dependent variable and show the regression estimates in Table A.14. Bonds with dual advisors have 1.03 to 1.06 fewer bids from nonadvisor underwriters from 2009 to 2011, significant at the 0.001% level in all specifications. Taking the advisor

²² These papers identify potential bidders in timber auctions as firms who bid in nearby auctions within a short amount of time as well as the observed participants in a given auction.

Table 4
Difference-in-differences model explaining number of bids submitted

	(1)	(2)	(3)	(4)	(5)
Dual advisor	−0.647 (0.166) .000	−0.643 (0.167) .000	−0.614 (0.169) .000	−0.616 (0.166) .000	−0.612 (0.166) .000
Dual advisor X Post	0.448 (0.087) .000	0.445 (0.087) .000	0.462 (0.088) .000	0.426 (0.086) .000	0.423 (0.086) .000
Observations	20,038	20,038	20,038	20,038	20,038
Mean auction participation	5.313	5.313	5.313	5.313	5.313
Year and issuer FE	Y	Y	Y	Y	Y
State economic and policy controls	Y	Y	Y	Y	Y
Market climate controls		Y	Y	Y	Y
Issuer type-by-year FE			Y	Y	Y
Size, refund, and call controls				Y	Y
Maturity terciles					Y

This table shows the estimates from regressions of primary market issue outcomes on type of advisor before and after the MSRB Rule G-23 reform for all competitive, general obligation, tax-exempt issues matched between the SDC Platinum database and Bond Buyer that employ financial advisor. The dependent variable is the total number of underwriters who submit bids in each competitive sale. All specifications control for year fixed effects issuer fixed effects, potential bidders, and state economic and policy controls. Column 2 adds controls for market conditions with SIFMA yields and 1- and 10-year swap spreads. Column 3 adds flexible trends for different types of issuers. The specification in column 4 adds controls for bond characteristics intrinsic to the project including size, refund status, and callability, while column 5, the preferred specification, adds fixed effect for years to maturity. See Section 3 for more information and discussion. Standard errors clustered at the issuer level are shown in parentheses with *p*-values below. Author's calculations using data from [SDC Platinum \(2016\)](#), [The Bond Buyer \(2016\)](#), and [Bergstresser and Luby \(2018\)](#).

out of the set of potential bidders increases participation by nonadvisors by 0.88 to 0.92 underwriters, which is also significant at the 0.001% level in all specifications and represents a 17.6% increase in the number of nonadvisors vying for underwriting business.

Figure 5 showcases an event study version of the preferred specification in column 5 of Table A.14 that is normalized such that the difference in 2011 equals zero. Visual examination of the pre-trend shows that, much like the interest cost regressions, there is no differential trend in auction participation for dual advisor issues before the MSRB Rule G-23 reform, but underwriter competition jumps by 0.5 nonadvisor underwriter bids immediately in 2012 and continues increasing in future years to 0.9 additional nonadvisor underwriters competing for underwriting business relative to the preperiod mean. Participation in auctions for municipal bond underwriting increases such that dual advisor issues and independent advisor issues have a very similar amount of competition after the reform, and this appears consistent across the distribution of participation shown in Figure 1.

3.3 Heterogeneity of Rule G-23 impacts

Municipal bonds differ widely in observable characteristics, the relative participation in the auctions, and in the potential for adverse selection. Broadly speaking, nonadvisor underwriters are informed about the state of the market, but may be less informed on the specifics of a certain bond—especially if the

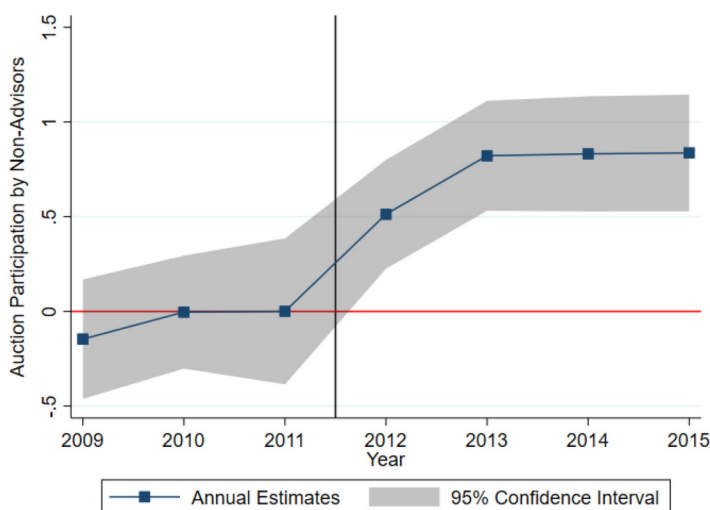


Figure 5
Treatment effect of dual advisor on number of nonadvisor auction participants

This figure reports the annual effects of having a dual advisor on the number of nonadvisor underwriters competing for underwriting business in competitive municipal bond sales as estimated by Equation (2). The graph is scaled such that the mean effect in the preperiod is zero. The specification reported in this figure corresponds to column 5 in Table 4 using all controls. See Section 3 for more information and discussion. Standard errors are clustered at the issuer level, and 95% confidence intervals are included. Author's calculations using data from SDC Platinum (2016), *The Bond Buyer* (2016), and Bergstresser and Luby (2018).

advisor or issuer does not readily provide that information. One margin of variation in publicly available information outside of the advisor's control is refunding status. One quarter of the competitive primary market is made up of issues that are refunding existing debt. Refunding issues, as opposed to new money issues, are used to refinance existing bonds, often at a lower interest rate or in a way that reduces short term cash outflows for municipalities. If there is a sense in which advisors are able to withhold some relevant information from other potential underwriters to accentuate adverse selection in the underwriting auctions, it should matter more for issues where other underwriters cannot already look to the secondary market to learn about the exact demand for a certain bond. Figure 6 replicates the baseline results separately for new money and refunding issues. For refunding issues, dual advisors are not associated with any borrowing cost change around the regulation. However, borrowing costs for new money issues decrease with a larger magnitude than in the baseline results. This heterogeneity is suggestive of dual advisors having some control over available information for new money issues in the primary market even though nonadvisor underwriters are informed about ongoing trading such as for refunding issues.

There are other ways of separating municipal bond issues based on expected participation or potentially exploitable asymmetric information. Garrett et al. (2023) show that competition from an additional potential underwriter has

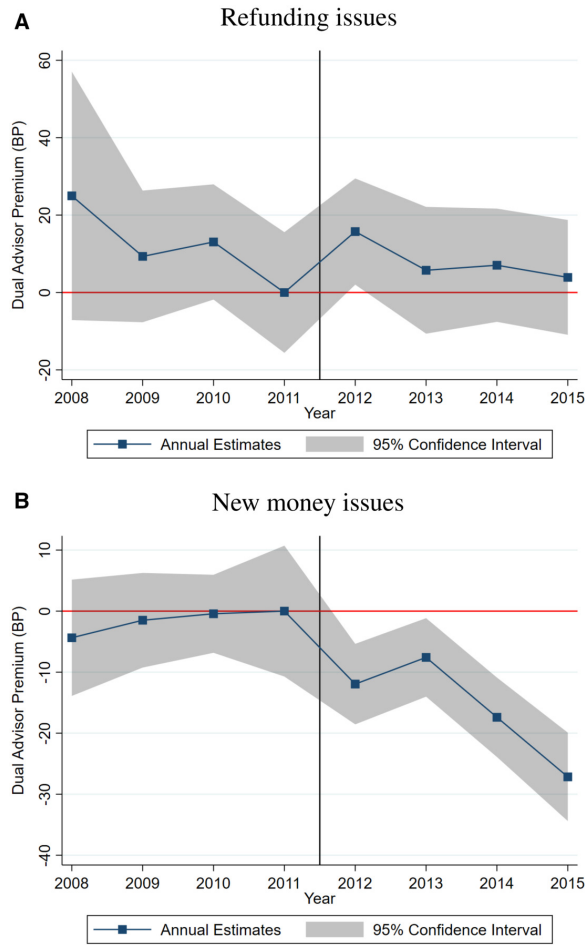


Figure 6
Treatment effect of dual advisor on winning bid (basis points), refunding heterogeneity

This figure reports the annual effects of having a dual advisor on TIC in basis points as estimated by Equation (2). The levels in the graphs are scaled such that the mean effect in 2011 is zero. The specification reported in this figure corresponds to column 5 in Table 1 using all controls. Panel A limits the sample to refunding issues that are being issued to finance payments on outstanding bonds. Panel B restricts the sample to new money issues. The lack of effect in panel A and the large negative effect in panel B is suggestive of the regulation encouraging advisors to provide more information on new money issues that the market is already providing for outstanding issues. See Section 3 for more information and discussion. Standard errors are clustered at the issuer level, and 95% confidence intervals are included. Author's calculations using data from SDC Platinum (2016), The Bond Buyer (2016), and Bergstresser and Luby (2018).

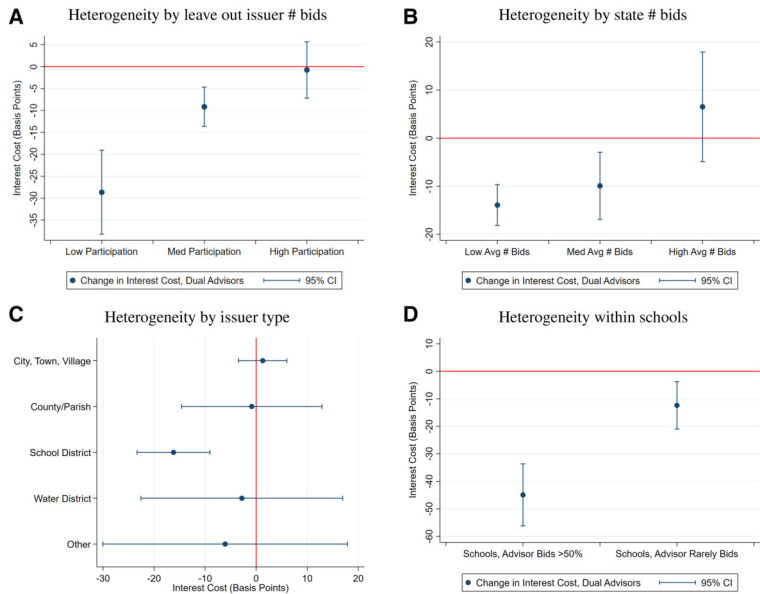


Figure 7
Estimates from regressions of bond issue outcomes on Dual advisor X Post, heterogeneity

This figure reports difference-in-differences estimates for borrowing costs as estimated by Equation (2). The specification reported in this figure corresponds to column 5 in Table 1. In panel A, low participation bonds are those bonds issued by issuers who have 4 or fewer underwriters submitting bids, medium participation is 4 to 6 bidders, and high participation is more than 6. The decrease in borrowing costs is driven by low participation issuers where the marginal increase in underwriter participation would have a larger effect on borrowing costs. Panel B shows the same pattern splitting states into thirds by the same metric. Panel C splits the treatment by issuer type and shows that school districts experience the largest impacts. Within Schools, the effect is concentrated among issuers who regularly received bids on their bonds from their advisors, with these previously reliant issuers seeing borrowing costs drop by 45 bps, which is shown in panel D. See Section 3.3 for more information and discussion. Standard errors are clustered at the issuer level, and 95% confidence intervals are included. Author's calculations using data from *SDC Platinum* (2016), *The Bond Buyer* (2016), and *Bergstresser and Luby* (2018).

the largest effect on borrowing costs in bond sales where there are relatively few other underwriter bidders. Rule G-23 removed a single bidder without replacement, so it should be expected that the competitive effect have the most detrimental impact in auctions with fewer bidders. I separate bond issues by expected level of auction participation to test for differential impacts in auctions where the marginal bidder may matter more. To define the expected level of participation, I reintroduce the concept of “leave-out competition,” or the average number of underwriter bids for other issues by the same issuer, and separate the sample into thirds based on this measure. The low participation issues come from issuers who have one to four underwriters competing for business, while the high participation issues come from issuers who have six or more underwriters bidding in their other auctions on average. This definition of participation for other issues minimizes the endogeneity of the advisor type choice from issue-specific idiosyncratic factors.

I estimate the difference-in-differences regression from Equation (2), while adding interactions for each third of the sample according to expected participation. Panel A of Figure 7 shows that the low participation issues are the ones for whom borrowing costs are decreasing, with an average 29-bps decrease in yields. High participation issues with dual advisors see an insignificant decrease in borrowing costs relative to similar bond auctions with independent advisors. The medium participation issues have a decline in interest cost that is statistically significant at the 1% level, but is smaller in magnitude than the impact for low participation issues. The declining impact of Rule G-23 as more participants are expected is not consistent with the missing bid from the dual advisor being the first-order impact of Rule G-23.

The same pattern holds when comparing across average state auction participation instead of the issue-specific expected participation from the leave-out measurement. Panel B of Figure 7 presents results of the test where participation is defined at the state level to take advantage of the Butler (2008) insight that municipal underwriting markets are largely local to the state. I sort states based on the average number of bids submitted across all auctions and then choose the first 16 states as the low category, the next 17 states as the medium category, and the final 17 states as the high category.²³ The states with relatively less attention in any given auction on average are the places where issuers with the largest impacts reside. The magnitude of estimates monotonically moves toward zero as there are more participants for each auction on average. Internet Appendix C.6 presents six additional splits across state auction characteristics and shows that the estimates are uniformly larger in states with more segmented markets and higher variance auction outcomes.

In panel C, I present a split based on issuer type instead of expected or observed participation in auctions. Different types of issuers access financial markets for a variety of reasons, and have differing levels of sophistication with which they approach the market. Entities ranging from states and state agencies to school districts all use auctions to find underwriters, and the advisor plays a different role and has more control over the issue depending on the issuer's needs. I separate issuers into five categories: (1) cities, towns, and villages, (2) counties and parishes, (3) school districts, (4) water districts, and (5) states, state agencies, other districts, and universities, which I refer to jointly as "other." Panel C of Figure 7 shows the difference-in-differences treatment effect split by issuer type categories. The decrease in borrowing costs is concentrated among dual advisor issues from school districts, which experience a decrease of 16 bps, while no other category has a statistically significant coefficient. Given the importance of municipal bond financing for public school buildings in much of the United States, school bonds make up 41% of the sample—the largest of all borrower types. School districts issue

²³ The 50 states are not perfectly divisible by three so this normalization is arbitrary, but the results are the same moving the rounding error to any group.

much less frequently than other entities, four times on average instead of 6 for the rest of the sample, and schools are much less likely to have different underwriters for different issues in the preperiod. School districts also pay higher interest costs than other municipalities. In the sample, school districts pay yields of 226 bps, while nonschool districts only pay yields of 210 bps on average.²⁴

Panel D of Figure 7 splits the schools further to examine those who regularly received bids by their advisor before rule G-23, which I define as more than half of their issues. The issues where the dual advisor regularly participated are those where both their missing bid could be most important, and also where the dual advisor could be using their role to increase profitability of their underwriting business.²⁵ Strikingly, the decrease in borrowing costs is four times larger for schools who previously had regular bids by their advisors. The treatment effect for this subsample is a decrease of 45 bps.

It is new money issues from low auction participation issuers and school districts who appeared reliant on advisor bids that are experiencing the decrease in borrowing costs. These are particularly opaque issues where the importance of adverse selection dominates the importance of an additional competitor. The following section explores the specific mechanisms that can explain why the adverse selection channel dominates and participation is increasing when a dual advisor is no longer allowed to underwrite municipal bonds.

4. Mechanism Evidence

The above analysis uses within-issuer variation to determine what happens to municipal bonds issued with dual advisors after those advisors are no longer allowed to act as the underwriter. Borrowing costs decrease by 11.4 bps, 5.3% of the average yield, driven by more aggressive auction participation from nonadvisor underwriters. This effect is largest for school districts that previously received lots of bids from their dual advisors, who experience a decrease in borrowing costs of 45 bps.

The question remains, why is removing the dual advisor from the underwriting auction effective at limiting the negative cost effect of the conflict of interest without limiting total auction participation? In the remainder of the paper, I enumerate additional evidence for two interrelated mechanisms

²⁴ While school districts do have a higher average yield to start with, Internet Appendix C and Figure A.8 show that the larger impact for schools is true in levels and also in relative terms as a share of interest cost.

²⁵ Another way of thinking of the variation by preperiod advisor auction participation is that different advisors have different propensities to use their advising arms to push business to their underwriter. In this case, we can define $Dual_i \in [0, 1]$ as the share of issues in which each advisor bid before 2011. A version of the difference-in-differences estimation with this definition of the treatment variable is included in Internet Appendix C.4 and shows that a dual advisor who bid in all of their issues was associated with a 21.5-bps decline in borrowing costs while the average dual advisor only saw a borrowing cost decline of 10.7 bps. This suggests that the municipality reliance on dual advisor bids in the preperiod could be a function of the advisor and not necessarily a specific characteristic of the issuer.

through which the regulation could affect the market and borrowing costs: (1) dual advisor bonds lack standardization in a way that may accentuate adverse selection benefiting themselves in 2008–2011 and (2) nonadvisor underwriters who outbid advisors for business in the preperiod have relatively lower gross spreads. The latter point is evidence of the winner’s curse. I then discuss how each affects the impact of such regulation aimed at an advisor conflict of interest.

4.1 Bond standardization, certification, and secondary market liquidity

[Schapiro \(2010\)](#) notes that the conflict of interest for dual advisor-underwriters may manifest in trying to raise interest costs, but that there are other margins on which an advisor can “guide the municipality towards securities tailored to his firm’s advantage.” Dual advisors who also have an interest in underwriting may advise a municipality toward debt that is more personalized for their own portfolios. Customized bonds are not inherently bad for an issuer if that customization helps the municipality either match payments to governmental cash flows or deal with other idiosyncratic local needs. However, customized bonds can make it difficult for investors to correctly judge the quality of the asset or to accurately price such bonds.²⁶

I estimate the change in bond structure on several margins for dual advisor issues around the update of MSRB Rule G-23 using the difference-in-differences regression described by Equation (2). First, I estimate changes in bond series characteristics that could directly affect the pricing: years to maturity, principal value, call provisions, presence of credit ratings, and credit enhancements. The estimates for these outcomes are shown in Table 5. Column 1 shows that dual advisor issues on the whole are decreasing the number of years to maturity, and further that the effect is concentrated among school districts. School district bonds with dual advisors have 1.18 more years to maturity in the preperiod, but the length decreases by 0.59 years on average in the postperiod, which is not statistically significant as a marginal impact relative to the rest of the dual-advised issues. However, a joint test of the coefficient for Dual Advisor $\times$ Post plus Dual Advisor $\times$ School $\times$ Post has a p -value of .044, which is marginally significant and so I am careful that it could be a margin of endogenous response although the effect is small and messy. Similarly, dual advisor issues are 5.8 percentage points (coefficient p -value = .033, joint test p -value = .014) more likely to have credit ratings

²⁶ The notion of complexity in the structure of a municipal bond has multiple accepted definitions in the literature. One strand of literature including [Brancaccio and Kang \(2022\)](#) defines the complexity of financial characteristics by the presence or amount of nonstandard covenants like sinking funds, call functions, variable coupon rates, or non-standard-interest frequency. Another literature studies textual complexity of disclosure. I focus on the former despite the potential difficulty in interpreting magnitudes. [Farrell et al. \(2023\)](#) develop a model of textual complexity where complexity can be valued differently across investor clienteles, which precludes me from making clear predictions about textual complexity in this scenario.

Table 5
Difference-in-differences model explaining bond characteristics

	Years to maturity	ln(Size)	Call	Rated	Enhanced
Dual advisor	−0.216 (0.593) .716	0.046 (0.052) .374	−0.016 (0.035) .641	0.021 (0.014) .157	0.026 (0.022) .240
Dual advisor X Post	−0.067 (0.302) .825	−0.005 (0.036) .892	0.022 (0.021) .291	−0.003 (0.014) .830	−0.031 (0.013) .015
Dual advisor X School	1.178 (0.886) .184	−0.118 (0.092) .204	0.017 (0.059) .774	−0.095 (0.040) .017	−0.237 (0.057) .000
Dual advisor X School X Post	−0.592 (0.439) .178	−0.055 (0.054) .309	−0.021 (0.029) .476	0.058 (0.027) .033	0.176 (0.027) .000
Observations	20,051	20,051	20,051	20,051	20,051
Dep. mean	10.804	1.877	0.589	0.709	0.169
Schools dep. mean	10.727	1.831	0.513	0.611	0.316
Issuer and year FE	Y	Y	Y	Y	Y
State economic and policy controls	Y	Y	Y	Y	Y
Size-by-year controls	Y		Y	Y	Y

This table shows the estimates from regressions of bond characteristics on type of advisor before and after the MSRB Rule G-23 reform for all competitive, general obligation, tax-exempt issues matched between the SDC Platinum database and Bond Buyer that employ financial advisor. The outcome in column 1 is the number of years to maturity. The outcome in column 2 is the natural logarithm of size in millions of dollars of par value. The outcome in column 3 is an indicator equal to one if a bond package has a call provision. The outcome in column 4 is an indicator equal to one for bonds with a credit rating from S&P, Moody's, or Fitch. The outcome in the fifth column is an indicator equal to one if a bond has insurance or a guarantee. The upper panel includes all issuers in the sample, while the lower panel only includes school districts. The dual advisor bonds become marginally more likely to be rated and more likely to use a credit enhancement. However, there is not a statistically significant change in the maturity or size of the bonds, or in the use of call provisions. See Section 4.1 for more information and discussion. Standard errors clustered at the issuer level are shown in parentheses with *p*-values below. Author's calculations using data from [SDC Platinum \(2016\)](#), [The Bond Buyer \(2016\)](#), and [Bergstresser and Luby \(2018\)](#).

after regulation²⁷ and also 17.6-percentage-points (*p*-value < .001) more likely to use a third-party credit enhancement, with both effects being concentrated exclusively in school districts. I find no evidence of changes in the average size of issues or in the use of call provisions. Changes in additional outcomes are shown and discussed in [Internet Appendix C.8](#).

Given the number of margins on which an advisor can make small changes to a bond issue that have effects on final investor demand, I also present a more general proxy for the general desirability of buying a bond: secondary market liquidity.²⁸ Liquidity is important in the municipal bond market where transactions with investors are done over-the-counter (OTC) by

²⁷ This credit rating result is consistent with findings from [Clarke \(1997\)](#), which finds dual advisor issues are less likely to have credit ratings in Texas.

²⁸ The interpretation of liquidity in this manner is more in line with the arguments presented in [Jankowitsch, Nashikkar, and Subrahmanyam \(2011\)](#) than in [Glosten and Milgrom \(1985\)](#). Asymmetric information among final investors is likely important in driving large differences in trading prices, but, in relation to the effects of MSRB Rule G-23, the asymmetry is driven by lack of standardization where some investors may have a difficult time pricing bonds. I am not assuming that some investors necessarily have additional information about each issue and that this information changes with the regulation.

registered broker-dealers. The nature of the OTC municipal bond market makes measuring liquidity a challenge because most bonds only trade once every few months after the first month. The market for a given asset is very thin, so bid-ask spreads, among other measures, are not directly available.

I follow [Jankowitsch, Nashikkar, and Subrahmanyam \(2011\)](#) to create an accessible measure of liquidity in the municipal bond market.²⁹ If a bond is easy to trade and desirable for investors, it is easier for broker-dealers to find another investor to purchase the bond in the case that one investor wants to sell. I use the [Municipal Securities Rulemaking Board \(2019\)](#) EMMA database to define average prices for investor purchases relative to investor sales.³⁰ On average in this sample, municipal bond investors pay \$1.30 more for each \$100 of par value to purchase a bond than they receive when they sell it back to the broker-dealer. I refer to this gap as the buy-sell spread.³¹

Regression estimates of the buy-sell spread on advisor type using the difference-in-differences design described in Equation (2) are shown in Table 6. These regressions are estimated at the bond level instead of the bond issue (auction) level in order to add controls for modified duration. Consistent with [Schwert \(2017\)](#), longer duration bonds are less liquid. Conditional on all bond observables, a 1-unit increase in modified duration increases the buy-sell spread by 6.3 cents. Notably, dual advisor issues are less liquid (higher buy-sell spread) before 2012, but such issues see a notable increase in liquidity after MSRB Rule G-23. The liquidity increase is comparable in effect size to decreasing the modified duration of each bond by 2.5 years. The buy-sell spread for dual advisor issues drops by 15.4 cents (*p*-value, .056) in the postreform period according to the preferred specification in column 5.

As a falsification test, I compare price dispersion for the dual-advised bonds treated by the regulation to the bonds issued by the same advisors in the preperiod. Overall, liquidity in the municipal bond market is increasing and price dispersion is decreasing during the sample. I compare the concurrent price dispersion in 2012–2015 for bonds issued with dual advisors in the preperiod to the price dispersion of dual advisor bonds issued after regulation and show the results in [Figure A.13](#). This comparison shows that the bonds issued with dual advisors have relatively less price dispersion than bonds issued from the same issuers with the same advisors in the preperiod that are trading at the same time. This gives further evidence that the observed change in price dispersion

²⁹ [Schestag, Schuster, and Uhrig-Homburg \(2016\)](#) and [Schwert \(2017\)](#) use variations of the same measurement concept.

³⁰ The average municipal bond in this sample trades less than 10 times in the first full year after issue, which precludes measurement of intraday price variation. The average prices paid by investors for municipal bonds are calculated for each bond for the first year of trading, omitting trades in the first month where prices vary widely. Average buying and selling prices for investor trades are pooled for the remaining 11 months.

³¹ The buy-sell spread is similar to the price dispersion measure included as part of the estimated liquidity spread in [Schwert \(2017\)](#) without including a consensus valuation around which dispersion is centered.

Table 6
Difference-in-differences model explaining buy-sell spread

	(1)	(2)	(3)	(4)	(5)
Dual advisor	0.190 (0.117) .103	0.185 (0.116) .113	0.200 (0.115) .082	0.170 (0.117) .144	0.165 (0.117) .158
Dual advisor X Post	−0.173 (0.080) .030	−0.188 (0.079) .017	−0.193 (0.081) .017	−0.156 (0.081) .054	−0.154 (0.081) .056
Average trade size ($\times 10^6$)	−0.074 (0.011) .000	−0.075 (0.011) .000	−0.076 (0.011) .000	−0.073 (0.011) .000	−0.072 (0.011) .000
Modified duration	0.066 (0.002) .000	0.065 (0.002) .000	0.065 (0.002) .000	0.064 (0.002) .000	0.063 (0.002) .000
Observations	43,544	43,544	43,544	43,544	43,544
Mean price gap	1.356	1.356	1.356	1.356	1.356
Year and issuer FE	Y	Y	Y	Y	Y
State economic and policy controls	Y	Y	Y	Y	Y
Market climate controls		Y	Y	Y	Y
Issuer type-by-year FE			Y	Y	Y
Size, refund, and call controls				Y	Y
Maturity terciles					Y

This table shows the estimates from regressions of secondary market issue outcomes on type of advisor before and after the MSRB Rule G-23 reform for all competitive, general obligation, tax-exempt issues matched between the SDC Platinum database and Bond Buyer that employ financial advisor. The dependent variable is the gap between what investors pay to buy bonds and the price investors receive when they sell bonds to registered broker-dealers in the over-the-counter municipal bond market in dollars per \$100 of par value. This gap is smaller for more liquid securities. Observations are defined at the CUSIP level, instead of the issue level. All specifications control for year fixed effects issuer fixed effects, and state economic and policy controls. Column 2 adds controls for market conditions with SIFMA yields and 1- and 10-year swap spreads. Column 3 adds flexible trends for different types of issuers. The specification in column 4 adds controls for bond characteristics intrinsic to the project including size, refund status, and callability, while column 5, the preferred specification, adds fixed effect for years to maturity. See Section 4.1 for more information and discussion. Standard errors clustered at the issuer level are shown in parentheses with *p*-values below. Author's calculations using data from SDC Platinum (2016), The Bond Buyer (2016), Bergstresser and Luby (2018), and Municipal Securities Rulemaking Board (2019).

for dual advisor issues is not explained by the decrease in price dispersion for bonds that are not directly affected by Rule G-23.

The increase in standardization and liquidity for dual advisor issues after regulation has two complementary effects on issuers: (1) borrowing costs decrease directly because secondary market investors are willing to pay more for bonds, and (2) nonadvisor underwriters participate in primary market auctions more actively because the bonds are be easier to sell.

4.2 Winner's curse and asymmetric information

Auctions with a bidder who has more (valuable) information than other the other participants in the auction about the value of the item being auctioned may not be revenue maximizing for the seller because of the large adverse selection effect of allowing such a participant (Hendricks, Porter, and Wilson 1994). If the dual advisor has some additional inside information about a bond issue they are advising that makes their assessment of market value for the bond better than other potential underwriters, then any other underwriter who wins the auction will have lower ex post profits. This is evidence of the winner's

course as a result of adverse selection. When nonadvisor underwriters are more likely to have lower profits if they win in an auction against a dual advisor, they are less likely to enter an auction at all and will bid less aggressively conditional on entering.

Underwriter profits are observable in the municipal bond market as underwriter spreads: the interest cost paid by the municipality to the underwriter minus the yield to final investors. As a simple test for asymmetric information in dual advisor bond auctions before the MSRB Rule G-23 reform, I estimate a regression of gross spreads on bond characteristics and whether a dual advisor bids in or wins the competitive sale:

$$\text{spread}_{ijt} = \alpha \text{dual}_i + \beta_1 \text{dual bids}_i + \beta_2 \text{dual wins}_i + \Gamma X_{ijt} + \varepsilon_{ijt}. \quad (3)$$

I estimate Equation (3) using competitive, general obligation tax exempt bond issues of more than \$1 million, where X_{ijt} includes year fixed effects, flexible fixed effects for years to maturity from sale and refund status, the natural logarithm of size interacted with years, credit rating fixed effects, financial market indices, fixed effects for use of funds, and flexible trends for different issuer types.³²

Of particular interest, β_1 is the differential effect on underwriter profits of winning an auction that an advisor bids in if the advisor does not win.³³ If this estimate is negative, it is indicative of lower profits if an underwriter wins an auction against potentially better-informed advisor. This is evidence of the winner's curse. $\beta_1 + \beta_2$ is the effect of an advisor bidding on and winning their own issue.

Estimates of Equation (3) are shown in Table 7. $\hat{\beta}_1$ is negative, and statistically significant at the 5% level in the preferred specification in column 4, while $\hat{\beta}_1 + \hat{\beta}_2$ is indistinguishable from zero. These results are suggestive of the winner's curse where nonadvisor underwriters face a disadvantage bidding against an informed advisor and will bid less aggressively and be less likely to enter auctions where they have to bid against advisors. Nonadvisor underwriters have a gross spread that is 3.5-bps lower if they win an auction against a dual advisor. This is representative of 6% decrease relative to the mean gross spread of 56.7 bps.³⁴

Having an informed bidder in an auction decreases seller revenue because other market participants receive lower profits in the case that they do win

³² Issuer fixed effects are omitted because there is insufficient identifying variation in whether an advisor bids and wins within the same issuer.

³³ Only dual advisors are able to bid in auctions they advise, so the dual bids_i indicator is a proper subset of the dual_i indicator.

³⁴ These estimates of gross spread are larger than traditional reports of gross spreads because I use the average yield in the first 7 days of trading as the secondary market yield instead of the "offering price," which is a regulatory construct that is not always very close to the price at which bonds are actually sold. Yields on the first trade or reported as the offering price are usually substantially higher than the average yield at which a bond is sold by the underwriter.

Table 7
Regression of gross spread on dual advisor bidding behavior, 2008–2011

	(1)	(2)	(3)	(4)
Dual advisor	1.607 (1.467) .273	1.656 (1.466) .259	1.314 (1.456) .367	1.409 (1.407) .317
Dual advisor bids	−1.955 (1.566) .212	−2.217 (1.569) .158	−2.815 (1.547) .069	−3.547 (1.485) .017
Dual advisor wins auction	4.989 (1.834) .007	5.354 (1.839) .004	5.153 (1.843) .005	4.627 (1.862) .013
Observations	8,422	8,422	8,422	8,422
Mean spread (BP)	56.721	56.721	56.721	56.721
Year, state, and maturity FE	Y	Y	Y	Y
State economic and policy controls	Y	Y	Y	Y
Market climate controls		Y	Y	Y
Issuer type-by-year FE			Y	Y
Size, refund, and call controls				Y

This table shows the estimates from regressions of secondary market issue outcomes on dual advisor bidding behavior before and after the MSRB Rule G-23 reform for all competitive, general obligation, tax-exempt issues matched between the SDC Platinum database and Bond Buyer that employ financial advisor. The dependent variable is the gross spread calculated using average yields in the first 7 days after the bond is issued. All specifications control for year fixed effects, state controls, and maturity fixed effects. Column 2 adds controls for market conditions. The specification in column 3 includes year-by-issuer type fixed effects to allow flexible time trends, while column 4, the preferred specification, adds controls for size-by-year, refund status, and call provisions. See Section 4.2 for more information and discussion. Standard errors clustered at the issuer level are shown in parentheses with *p*-values below. Author's calculations using data from [SDC Platinum \(2016\)](#), [The Bond Buyer \(2016\)](#), [Bergstresser and Luby \(2018\)](#), and [Municipal Securities Rulemaking Board \(2019\)](#).

the auction against the informed bidder. In the case of the dual advisors, they receive higher profits when they win an auction than nonadvisors get if they win. Such asymmetric information deters the entry of nonadvisors, which is consistent with the results presented in Section 3. After the informed advisor is taken out of the auction, the other potential underwriters are more likely to enter the auction and compete. Further, [Internet Appendix C.4](#) shows that all participating underwriters, including the losing underwriters, bid more aggressively for dual advisor bonds after the regulation.

5. Potential for Net Benefits of Reform

The reform of MSRB Rule G-23 in November of 2011 is associated with a decrease in interest costs for municipalities and with more participation in primary market auctions. However, the interest cost benefits are not costless. A major remaining question is whether the associated costs are larger than the decrease in borrowing costs. Issuers could be worse off in three major ways given the interest cost decreases estimated above.

First, issuers hiring dual advisors may have valued the flexibility and nonstandard characteristics of their issues with dual advisors before 2011. It may be that issuers have a large willingness to pay for other characteristics like lining up specific expenditures with specific revenues that dual advisors had

more flexibility to offer when they could also serve as an underwriter of last resort in the auction stage. Quantifying the loss associated with less flexibility due to increased standardization is beyond the scope of this paper.

Second, dual advisors may increase the price of advising services after they can no longer be compensated in the second stage as underwriters. If the price of dual advising increases but issuers are relatively unable to substitute to other advisors whose costs do not increase, those municipalities that keep hiring dual advisors could be worse off.

Third, obtaining credit ratings or other third-party certifications and enhancements definitely adds an additional cost to issuance. Data on the costs of third-party credit certifications in the municipal bond market are notoriously opaque and are not available for individual bonds in general (Cornaggia, Cornaggia, and Israelsen 2018). However, it is possible to back out the average costs of different instruments on public filings from credit rating agencies and insurers.

While measuring price changes of advisory services and welfare losses due to standardization are beyond the scope of this paper, I can give an approximation of potential benefits net of these considerations by comparing the cost of credit ratings and enhancements to the change in borrowing costs. Joffe (2017) provides a review of the data that are available from public sources regarding the costs of credit ratings and credit enhancements.³⁵ On average, credit rating agency revenues imply that the cost of credit ratings for municipalities are the equivalent of about 1 bp in additional interest costs. Further, in 2014 Moody's published a list of prices for municipal bond credit ratings that spanned from \$7.5k to \$498k for the municipality. For very short term issues and very small issues, the costs of buying a credit rating are the largest proportionally because it is a fixed cost. An issuer paying \$7.5k for a credit rating for a 10-year, million dollar bond with a 3% yield to maturity is the equivalent of adding 8.8 bps onto the interest cost, which is over 8 times larger than the average cost of a credit rating, but still smaller than the decrease in borrowing costs for the whole sample on average, but only a small subset of issues actually acquire a new credit rating.

Credit enhancements are more expensive than credit ratings and, according to Table 5, 17.6% of school districts with dual advisors add some sort of credit enhancement after regulation relative to issues with independent advisors. On average, Joffe (2017) notes that such enhancements are priced around 1% of par value on average based on aggregate statistics. For the 10-year, million dollar bond with a 3% TIC above where fixed costs are expected to be most substantial, such a contract is the same as adding 11.8 bps onto the interest cost. For the set of issuers for whom the regulation appears to have some bite, namely, school districts who appeared reliant on bids from their dual advisors in

³⁵ In the muni market, credit rating agencies make money from two sources: (1) initial payments from municipalities and (2) subscription services from investors.

the preperiod, true interest costs decrease by 45 bps, so the subset that acquire credit enhancements on the margin would need to pay four times the price above for credit enhancements for this policy to have a net negative effect on their borrowing cost, but only a subset of these issuers have to acquire credit enhancements.

The school districts who acquire credit ratings and enhancements for the first time after their advisor is no longer allowed to underwrite do not appear to be systematically less creditworthy than the independently advised issuers who already used credit ratings and enhancements. I identify 191 school districts who exclusively employ dual advisors and never had credit ratings before November 2011 and graph the relative frequencies of credit ratings after November 2011 in panel A of Figure 8. The median credit rating for newly rated schools with dual advisors is Aa2 by Moody's rating scale, which is the same as the population of school districts and the total population of issuers. While limiting to a subset of 73 school districts who also never buy a credit enhancement during the sample in panel B, the newly rated schools are very similar in underlying credit worthiness to other school districts and nonschool borrowers without credit enhancements during this period.

Prohibiting advisors from being able to underwrite adds some new costs in terms of acquiring third-party credit certifications that increase available information to other parties in the market. The information revealed by these certifications suggests that the effect of regulation is not a marginal increase in credit worthiness, but that creditworthy borrowers were not successfully sharing that information with the market. These observable costs are smaller in net present value terms than the benefit received in lower borrowing costs for affected municipalities, but this hypothetical assumes that dual advisor prices are not changing and that issuers did not value the nonstandard characteristics of their bonds before regulation.

Another caveat to this exercise in comparing some costs and benefits is that it does not address the long-run outcomes of the advising or underwriting markets, so a total and long-run impact of the new Rule G-23 on welfare could be negative if it causes enough firms to leave the advising or underwriting businesses. As some descriptive evidence that this is not happening, I show the pattern of participating advisors and underwriters in the market from 2008 to 2015, which are increasing and flat, respectively, in Figure A.3.

6. Conclusion

Municipal financial transactions are not immune from the conflicts of interest that are pervasive in other markets, and they provide a novel window into how information asymmetries can be propagated by intermediaries to create additional adverse selection. By studying the response to the reformulation of MSRB Rule G-23 after Dodd-Frank, this paper shows that financial advisors who also offer underwriting services have a conflict of interest that raised



Figure 8
New Moody’s credit rating distribution, 2012–2015

This figure reports the distribution of credit ratings separately for those school districts with dual advisors who acquire ratings for the first time after 2011, all school districts without dual advisors, and all issuers without dual advisors. Panel A includes issuers who ever purchase a credit enhancement (including insurance, guarantees, or letters of credit), while panel B restricts to issuers who never purchase any sort of credit enhancement. Both figures highlight that the newly rated dual advisor bonds after MSRB Rule G-23 are not systematically rated lower than similar independently advised issuers, which suggests these borrowers were similarly creditworthy while issuing unrated debt in previous years. Author’s calculations using data from [SDC Platinum \(2016\)](#) and [Bergstresser and Luby \(2018\)](#).

interest costs for municipalities in the absence of regulation. Removing the ability to both advise and underwrite the same municipal bond issue decreases municipal borrowing costs by decreasing asymmetric information and adverse selection, which winds up increasing total auction participation by potential underwriters.

Municipal financial advisors have wide breadth to affect the borrowing outcomes for cities and states in the United States. Before Dodd-Frank and MSRB Rule G-23, municipal advisors who could also underwrite bonds give advice on issues that are less standard and less liquid in secondary markets, increase asymmetric information that benefits themselves, and scare away other potential underwriters.

These results also inform our understanding of conflicts of interest in other markets. Restricting the ability of advisors to underwrite changes the design of bonds dual advisors create as shown through increased standardization, sharing of inside information, and liquidity in secondary markets. This is consistent with a change in the advice that previously conflicted dual advisors are offering. Also, the fear of negative consequences regarding decreasing competition described in *Bond Dealers of America* (2019) and supported by existing literature in other markets does not manifest here. Allowing advisors to operate in both the advising and underwriting markets decreases participation in the underwriting market by deterring other underwriters from competing for business with an agent who can actively control the available information in the market and create adverse selection. Removing the advisor from the primary market increases participation from other underwriters and drives down interest costs for municipalities, especially those municipalities that are most opaque and where creating adverse selection is most possible.

Code Availability: The replication code is available in the Harvard Data-verse at <https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/INIZAQ>.

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